GG-D-600D
August 18, 1982
SUPERSEDING
Fed. Spec. GG-D-600C
April 17, 1974

FEDERAL SPECIFICATION

DRAFTING INSTRUMENT SETS:

AND DRAFTING INSTRUMENTS

This specification was approved by the Commissioner, Federal Supply Services, General Services Administration for use by all Federal Agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers four drafting instrument sets, and individual drafting instruments.

1.2 Classification. Drafting instruments shall be supplied in the following sets, as specified (see 6.2):

Set 1 - Field set
Set 2 - Office Set
Set 3 - Pocket set
Set 4 - Mini set

When required, individual drafting instruments shall be as specified (see 6.2).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

FSC 6675

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Federal Specifications:

GG-C-871/1	- Curves, French (Irregular), Set or Individual.
GG-P-125	- Pen, Ruling, Draftsman's, Contour, Swivel Action.
GG-P-133	- Pen, Ruling, Draftsman's, Rail-Road, Swivel Blade, 5-Inch.
GG-P-681	- Protractor, Circular and Semi-circular, (Mils, Degrees).
GG-T-671/1	- Triangles, Drafting, Plastic, Transparent.
QQ-P-35	- Passivation Treatments for Corrosion-Resisting Steel.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-B-640	- Boxes, Fiberboard, Corrugated, Triple-Wall.

Federal Standards:

FED. STD. No. 123	- Marking for Domestic Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal specifications, standards, and commercial item descriptions, as outlined under General Information in the Index of Federal Specifications, Standards, and Commercial Item Descriptions. The Index, which includes cumulative bimonthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification, other Federal Specifications, and Commercial Item Descriptions required by activities outside the Federal Government for bidding purposes are available without charge from General Services Administration Business Service Centers in Boston; New York; Philadelphia; Washington, DC; Atlanta; Chicago; Kansas City, MO; Fort Worth; Houston; Denver; San Francisco; Los Angeles; and Seattle, WA.

(Federal Government activities may obtain copies of Federal specifications, standards, commercial item descriptions, and the Index of Federal Specifications, Standards, and Commercial Item Descriptions from established distribution points in their agencies.)

Military Specification:

MIL-P-116	- Preservation, Methods of.

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Military Standards:

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.

(Copies of Military specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/ASTM B 36 - Brass Plate, Sheet, Strip, and Rolled Bar.
ANSI/ASTM B 211 - Aluminum-Alloy, Bar, Rod, and Wire.
ANSI/ASTM B 455 - Copper-Zinc-Lead Alloy (Leaded Brass) Extruded Shapes.

(Application for copies should be addressed to American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, AGENT

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., ATTN: Traffic Department, 1616 P Street, NW, Washington, DC 20036.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, ATTN: Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Description. The drafting instrument sets shall consist of the various instruments specified in table I contained in a fitted case. Individual instruments may also be acquired using this specification.

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TABLE I. Drafting instrument sets.

Description	Set 1 field	Set 2 office	Set 3 pocket	Set 4 mini	Reference
Case	1	1	1	1	3.7
Ruling pen, small	1	1		1	3.8.3
large	1	1	1		3.8.4
Bow compass, pen	1	1			3.9.1.1
pencil	1	1			3.9.1.1.2
combination				1	3.9.1.1.3
large				1	3.9.1.2
Compass, large, friction	1	1	1		3.9.2
Lengthening bar	1	1			3.9.2.1
Dividers, small	1	1			3.10.1
large	1	1		1	3.10.2
Center horn		1			3.11
Spare parts and lead holder	1	1	1	1	3.12
Screw driver (when required)	1	1	1	1	3.13
Protractor, type I, class A, 8-Inch		1			GG-P-681
Triangle, class A, 30° -60° 6-inch		1			GG-T-671/1
class B, 45° 4-inch		1			GG-T-671/1
Ruling pen, contour		1			GG-P-125
railroad		1			GG-P-133
Curve, type I, pattern 1		1			GG-C-871/
type I, pattern 16		1			GG-C-871/

3.2 First article. When specified (see 6.2), the contractor shall furnish three or more sample units for first article inspection and approval (see 6.3).

3.3 Interchangeablilty. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. Pencil leads and needles used in the compasses and dividers shall be interchangeable.

3.4 Material. The instruments shall be corrosion-resisting steel, or chrome or nickel-plated brass at the option of the contractor, except that when a specific material is required, the material shall be as specified (see 6.2). Material not specified shall be selected by the contractor and shall be subject to all provisions specified herein (see 6.4 and 6.5).

3.4.1 Corrosion-resisting steel. Corrosion-resisting steel shall be passivated in accordance with QQ-P-35, and shall conform to standard ANSI designations for alloys S30100, S30400, S42000, or S43000, condition and temper as required.

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3.4.2 Chrome or nickel-plated brass. Brass parts shall be of brass conforming to ANSI/ASTM B 36 C26800, temper H02, and shall be chrome-plated to a thickness of not less than 0.004 inch or nickel-plated to a thickness of not less than 0.010 inch.

3.4.3 Aluminum. Aluminum used for handles, lengthening bars, and other parts shall conform to ANSI/ASTM B 211, alloys A92011 or A92024.

3.5 Construction. Instruments may be constructed of solid material, truss type, or hollow tube. If construction is truss type, each truss shall be not less than 0.060 inch thick. If hollow tube type, the wall thickness shall be not less than 0.045 inch throughout the length of the piece. All compass and divider points shall be aligned to register accurately in closed position. Attachments, such as lengthening bars or beams, and pen and pencil parts, shall fit freely; and, when tightened, shall not twist or turn.

3.6 Handles. Compass and divider handles shall be the same material as the instrument, and shall be finely knurled or shall have fine longitudinal corrugations. Handles shall be between 0.625 and 0.875-inch long and between 0.187 and 0.312-inch in diameter.

3.6.1 Screws. All screws and screw studs shall be corrosion-resisting steel. Screw heads and adjusting nuts shall be knurled and shall be leaded brass conforming to ANSI/ASTM B 455, alloy C38500, or corrosion-resisting steel specified in 3.4.1.

3.6.2 Needle points. Needle points shall be chrome-plated steel or corrosion-resisting-steel. Needles shall have a shoulder point on one end and a needle point on the other end. Each point shall be concentric with the body of the needle and shall be sharp.

3.6.3 Pencil leads. Each instrument requiring pencil leads shall have a pencil lead firmly clamped in its proper socket. Pencil leads shall be between 1- and 1-1/4 inches long and 0.078 inch, plus 0.002 inch or minus 0.005 inch thick. Pencil lead shall have a 4H degree of hardness and shall have a chisel point.

3.7 Cases.

3.7.1 Sets. Each set of drafting instruments shall be contained in a fitted case with a hinged lid. The case shall be wood, plastic or aluminum. The bottom of each case shall have recesses shaped to fit the instrument for which it is intended. When in the case, no instrument shall come in contact with any other instrument or attachment in the case. The inside of the case, including that part that covers the instruments when the case is closed, shall be lined with dark colored velvet, velveteen, or plastic foam rubber. Fasteners shall be provided to keep the case closed when not in use. Fasteners and hinges shall be corrosion-resisting metal.

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3.7.2 Individual instruments. When individual instruments are furnished, each instrument shall be supplied with a plastic case with a hinged lid and fastener, size fitted to the instrument; or a plastic vinyl pouch with fold over tucked flap.

3.8 Ruling pens (except contour and railroad pens).

3.8.1 Blades. Pen blades shall be spring-blade, spring-hinge, or cross-hinge type. The blade points shall have a hardness of 47, plus or minus 7, on the Rockwell C-scale. The blades shall be sharpened and set, and shall be adjustable to permit drawing clean, uniform, unbroken lines between 0.006-inch and 0.062-inch in width on drawing paper, tracing cloth, and on matte-surfaced plastic sheet. The points shall not cut into the drafting medium under normal use.

3.8.2 Handles. Ruling pen handles shall be rigid plastic or metal, and shall be between 0.187 and 0.25-inch in diameter.

3.8.3 Small ruling pen. The small ruling pen shall be between 4-1/2 and 5-1/2 inches long. The blade portion shall be 2 inches, plus or minus 0.25 inch long.

3.8.4 Large ruling pen. The large ruling pen shall be between 5-1/2 and 6-1/2 inches long. The blade portion shall be 2-1/2 inches, plus or minus 0.25 inch long.

3.9 Compasses.

3.9.1 Bow compass. The opening and closing of the legs of the bow compass shall be controlled by a center adjusting thumb screw. The adjusting screw shall open and close the spacing firmly and smoothly and shall hold the legs rigidly while drawing circles.

3.9.1.1 Small bow compass. The small bow compass shall be between 3-1/2 and 4-1/2 inches long, and shall permit drawing circles from 0.031 inch to not less than 2 inches in diameter.

3.9.1.1.1 Compass pen. The compass pen shall be provided with a removable and adjustable needle in one leg and a ruling pen in the other leg. The pen blades shall be as specified in 3.8.1, except that the maximum width of the line shall be 0.050 inch.

3.9.1.1.2 Compass pencil. The compass pencil shall be provided with a removable and adjustable needle in one leg and a pencil lead holder on the other leg. The pencil lead holder shall accommodate the pencil lead specified in 3.6.3.

3.9.1.1.3 Combination. The combination bow compass shall be provided with a removable and adjustable needle in one leg and the other leg shall have an interchangeable pen holder and pencil holder. The pen holder shall have a pen as specified in 3.9.1.1.1 and the pencil holder shall accommodate pencil leads specified in 3.6.3. The pen holder and the pencil holder shall be held firmly in place without movement when tightened.

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3.9.1.2 Large bow compass. The large bow compass shall be between 6 and 6-1/2 inches long, and shall permit drawing circles from 0.062 inch to not less than 5 inches in diameter. One leg of the compass shall be provided with a removable and adjustable needle and the other leg shall have an interchangeable pen holder and pencil holder. The pen holder or the pencil holder shall be held firmly in place without movement when tightened. The pen holder portion shall have pen blades conforming to 3.8.1. The pencil portion shall accommodate leads conforming to 3.6.3.

3.9.2 Large friction compass. The large friction compass shall be between 6 and 6-1/2 inches long, and shall permit drawing circles from 0.062 to not less than 5 inches in diameter. The friction shall be sufficient to hold the dividers in any position during normal use. One leg of the compass shall be provided with a removable and adjustable needle and the other leg shall have an interchangeable pen holder and pencil holder. The pen holder or the pencil holder shall be held firmly in place without movement when tightened. The pen holder portion shall have pen blades conforming to 3.8.1. The pencil portion shall accommodate leads conforming to 3.6.3.

3.9.2.1 Lengthening bar. A lengthening bar or beam not less than 3-1/2 inches long shall be provided for use with the large compass to permit drawing larger circles than are normally possible with the compass alone. The lengthening bar or beam shall fit into the socket for the pen or pencil attachment, and the pen or pencil attachment shall fit into the lengthening bar or beam.

3.10 Dividers.

3.10.1 Small divider. The small divider shall be similar to the small bow compass specified in 3.9.1 and 3.9.1.1, except that both legs shall have adjustable and replaceable needle points as specified in 3.6.2.

3.10.2 Large divider. The large divider shall be friction type and shall be between 6- and 6-1/2 inches long. Both legs shall be equipped with adjustable and replaceable needles as specified in 3.6.2. The range, or spacing between the points shall be continously variable from less than 0.062 inch to more than 2-1/2 inches. Movement shall be free without binding or tight spots. The friction shall be sufficient to hold the dividers in any position during normal use.

3.11 Center horn. The center horn shall be a flat disc of transparent plastic mounted in a metal or plastic rim. The plastic disc shall be between 0.50 and 0.625 inch in diameter. The rim shall be equipped with three equally spaced needle-sharp pins protruding between 0.046 and 0.078 inch below the lower surface of the metal rim.

3.12 Additional parts and pencil lead holders. Additional parts and pencil lead holders may be in two separate holders or they may be combined into a single holder. Each holder shall be provided with a friction or threaded type removable cap. The holder(s) shall contain not less than the following items:

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3 Pencil leads (see 3.6.3).
3 Needle points (see 3.6.2)
2 Adjusting T-bolts and nuts (when required).
2 Eye bolts and clamp nuts (when required).

3.13 Screwdriver. When required for making adjustment on any instrument in any set, a screwdriver of the proper size shall be furnished with the set.

3.14 Identification marking for DOD only. The case of each set shall be identified in accordance with MIL-STD-130. Individual instruments need not be marked.

3.15 Workmanship. Materials shall be free from flaws or defects which could impair the function or appearance of the instruments. Surfaces shall be smooth, free from burrs and sharp corners (except needles), and shall be polished or have a satin finish.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

4.2 Classification of inspections. Inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4)

c. Examination of preparation for delivery (see 4.6).

4.3 First article inspection.

4.3.1 Examination. The first article instruments shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The first article instruments shall be tested as specified in 4.5.2. Failure of any test shall be cause for rejection.

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4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105, inspection level III.

4.4.2 Examination.

4.4.2.1 Samples. Samples selected in accordance with 4.4.1 shall be examined for the defects specified in 4.5.1. Acceptable Quality Level (AQL) shall be 0.4 percent defective for major defects and 2.5 percent defective for minor defects.

4.4.3 Tests. Samples selected in accordance with 4.4.1 shall be tested as specified in 4.5.2. AQL shall be 4.0 percent defective.

4.5 Inspection procedure.

4.5.1 Examination. The instrument sets and individual instruments shall be examined as specified herein for the following defects:

Major

101. The instruments contained in a set not as specified in table I.
102. Compass and divider points not alined.
103. Case not as specified for sets and individual instruments.
104. Material not as specified.
105. Dimensions of individual instruments not within specified tolerances.
106. Parts missing or broken.
107. Additional parts and pencil leads missing, or not as specified.
108. Additional parts and pencil lead holder(s) not as specified.
109. Attachments loose in the sockets, not adjustable.
110. Screwdriver missing (when required).
111. Instruments not finished as specified.
112. Workmanship not as specified.

Minor

201. Identification marking missing, incomplete, or illegible.

4.5.2 Tests.

4.5.2.1 Ruling pens. The adjusting screw of each pen shall be moved to determine freedom of movement and proper adjustment of the pen nibs. Each pen shall be filled with a drawing ink and tested by drawing lines on drawing paper, tracing paper, and matte surfaced plastic sheet. Failure to draw clean, uniform, unbroken lines at any thickness between 0.006-inch and 0.062-inch, or cutting into the drafting medium, shall constitute failure of this test. Prior to the test the drawing paper, tracing paper, or plastic may be cleaned.

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4.5.2.2 Compass operation. The adjusting screw of each compass and the ruling pen adjusting screw of the pen compass shall be moved to determine freedom of movement. Circles shall then be drawn with both pen and pencil compasses. Failure to draw clean, uniform, unbroken circles of radii specified in 3.9.1.1 and 3.9.1.2 for the small and large bow compass, respectively, shall constitute failure of this test. Failure of the pen compasses to draw circles with lines between 0.006-inch and 0.050-inch wide for the small compass and between 0.006-inch and 0.062-inch for the large compass also shall constitute failure of this test.

4.5.2.3 Friction test. The following test shall be conducted on friction-type dividers and compasses. The instrument shall be in closed position so that the points just touch. The instrument shall then be extended to full open position with one continuous movement, and then moved to the closed position. This constitutes one cycle. This cycle shall be repeated 25 times. Evidence of any of the following shall constitute failure of this test:

a. Rough spots or sticking during operation.

b. Evidence of abrasion or wear.

c. Loosening of parts.

d. Noticeable lessening of friction.

e. Nonalinement of points.

4.6 Examination of preparation for delivery. An examination shall be made to determine compliance with the requirements of section 5. The sample unit shall be one shipping container fully prepared for delivery. Sampling shall be in accordance with MIL-STD-105. The inspection level shall be S-2 with an AQL of 4.0.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A, B, or commercial as specified (see 6.2).

5.1.1 Level A. Each instrument in its case, and each drafting instrument set contained in its case shall be preserved in accordance with MIL-P-116, method IC-1.

5.1.1.1 Intermediate packaging. Individual instruments of like description, preserved as specified in 5.1.1, shall be intermediate packaged together in quantities of 25 in close-fitting boxes conforming to PPP-B-636, W6c. Instrument sets of like description preserved as specified in 5.1.1 shall be intermediate packaged together as specified for individual instruments except in quantities of 10. After closure, the boxes shall be waterproof-sealed method V in accordance with appendix to the box specification.

5.1.2 Level B, civil agencies. Individual instruments and instrument sets shall be preserved and intermediate packaged as specified for level A, except fiberboard boxes for intermediate packaging shall be domestic class. The boxes shall be closed in accordance with the appendix to the box specification.

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5.1.3 Commercial. Individual instruments and instrument sets shall be preserved and packaged in accordance with normal commercial practice. The package shall be designed to protect the contents against damage during shipment, handling, and storage.

5.2 Packing. Packing shall be level A, B, or commercial, as specified (see 6.2).

5.2.1 Level A. The drafting instruments and drafting instrument sets preserved and packaged as specified in 5.1 shall be packed in close-fitting boxes conforming to PPP-B-601, overseas type, style optional. Box closure and strapping shall be in accordance with the appendix to the box specification.

5.2.2 Level B. The drafting instruments and drafting instrument sets preserved and packaged as specified in 5.1 shall be packed in close-fitting boxes, conforming to PPP-B-636, class weather resistant, or PPP-B-640, class 2. Fiberboard boxes shall be waterproof-sealed in accordance with the appendix to the applicable box specification.

5.2.3 Commercial. The drafting instruments and drafting instrument sets preserved and packaged as specified in 5.1 shall be packed to assure carrier acceptance and safe delivery to destination at the lowest rates in fiberboard containers complying with Uniform Freight Classification rules or with National Motor Freight Classification rules.

5.3 Marking.

5.3.1 Military agencies. In addition to any special marking required in the contract or purchase order, packages and shipping containers shall be marked in accordance with MIL-STD-129.

5.3.2 Civil agencies. In addition to any special marking required in the contract or purchase order, packages and shipping containers shall be marked in accordance with FED. STD. No. 123.

6. NOTES

6.1 Intended use. Drafting instruments are intended for use by architects, engineers, scientists and draftsmen for general drafting practice.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Set of drafting instruments required (see 1.2).

c. The individual instruments required (see 1.2).

d. When a first article is required for inspection and approval, and number of units required (see 3.2).

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e. Material required, when a specific material is required (see 3.4).

f. Level of preservation and packaging, and level or packing required (see 5.1 and 5.2).

6.3 First article. When a first article is required, it shall be tested and approved under the appropriate provisions of 7-104.55 of the Defense Acquisition Regulation (DAR). The first article should be preproduction instrument. The contracting officer should include specific instructions in all acquisition instruments regarding arrangements for examinations, test and approval of the first article.

6.4 Recycled material. It is encouraged that recycled material be used, when practical, as long as it meets the requirements of this specification (see 3.4).

6.5 Materiel deterioration prevention and control (MADPAC). The contracting officer should consider for inclusion in all applicable invitations for bid, requests for proposal, and contracts the following statement:

"The technical proposal shall provide assurance that all system/item parts shall be fabricated from compatible materials, suitably finished, to provide protection from corrosion and other forms of materiel deterioration. The assurance shall be in the form of a finish plan, utilizing state-of-art technology, which clearly identifies the specific finish or treatment to be used on the various forms of materiel deterioration in the applicable operating and storage environments to which they may be exposed."

MILITARY INTERESTS:

Custodians
Army - ME
Air Force - 99

Review activity
DLA-GS

User activities
Navy - MC, YD

CIVIL AGENCY COORDINATING ACTIVITIES:

GSA-FSS
VA-OSS
COM-NBS

Preparing activity: Army - ME

Project 6675-0716

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